Advanced micro lab

Lab report 3

Newton-Raphson 16-bit Implementation

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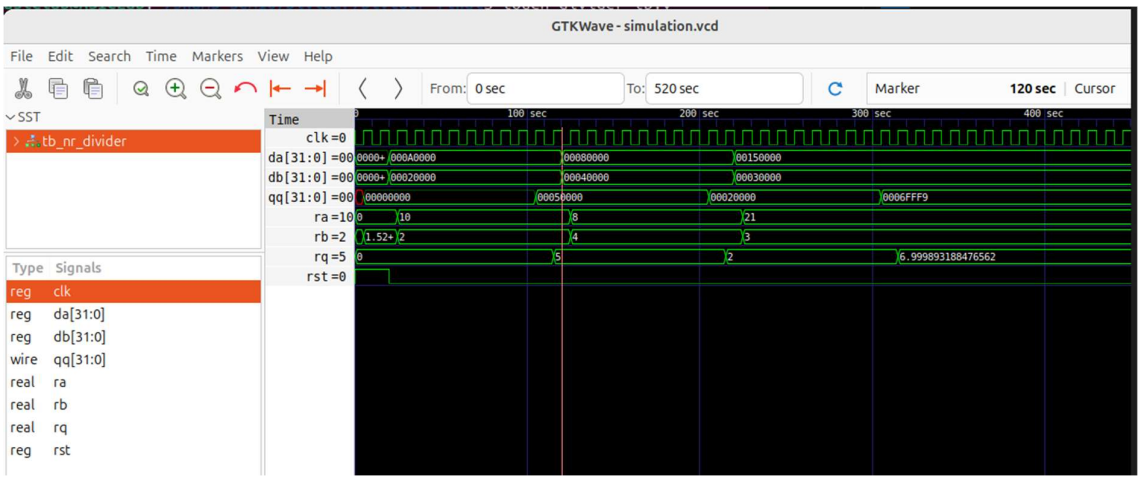
Task 1: Implementing the Divider

1. use Icarus

```
asiclab@ASICLab:~/shams_sem10/Divider/Divider_final$ vvp sim.vvp
VCD info: dumpfile simulation.vcd opened for output.
10 / 2 = 5.000000
8 / 4 = 2.000000
21 / 3 = 6.999893
divider_tb.v:45: $finish called at 520 (1s)
```

2. GTKWave



3. **<u>Run Yosys synthesis 3 times:</u>**

FPGA

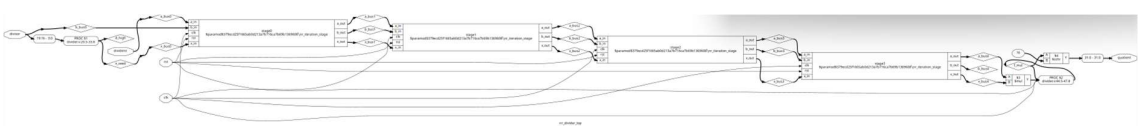
Before optimization

```
=== design hierarchy ===

   nr_divider_top                    1
     $paramod$379ecd25f1665ab0d213a7b716ca7b69b1369608\nr_iteration_stage

   Number of wires:                268
   Number of wire bits:           9013
   Number of public wires:          79
   Number of public wire bits:    2334
   Number of ports:                 37
   Number of port bits:            874
   Number of memories:               0
   Number of memory bits:            0
   Number of processes:              0
   Number of cells:               1635
     BUFG                            1
     CARRY4                        140
     DSP48E1                        40
     FDRE                          808
     IBUF                           66
     INV                           188
     LUT2                          343
     LUT4                           17
     OBUF                           32

   Estimated number of LCs:        180
```

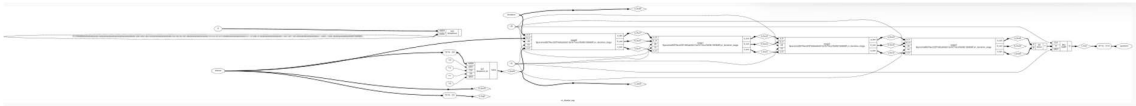


After optimization

```
nr_divider_top                    1
  $paramod$379ecd25f1665ab0d213a7b716ca7b69b1369608\nr_iteration_stage

Number of wires:                250
Number of wire bits:           8670
Number of public wires:          79
Number of public wire bits:    2319
Number of ports:                 37
Number of port bits:            874
Number of memories:               0
Number of memory bits:            0
Number of processes:              0
Number of cells:               1635
  BUFG                            1
  CARRY4                        140
  DSP48E1                        40
  FDRE                          808
  IBUF                           66
  INV                           188
  LUT2                          343
  LUT4                           17
  OBUF                           32

Estimated number of LCs:        180
```



Timing Report

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	84.996 ns	Worst Hold Slack (WHS):	0.152 ns	Worst Pulse Width Slack (WPWS):	49.020 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	1407	Total Number of Endpoints:	1407	Total Number of Endpoints:	335

All user specified timing constraints are met.

Report

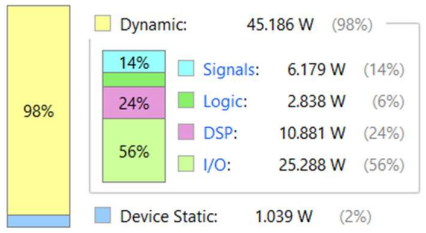
Summary

Power estimation from Synthesized netlist. Activity derived from constraints files, simulation files or vectorless analysis. Note: these early estimates can change after implementation.

Total On-Chip Power:	**46.226 W (Junction temp exceeded!)**
Design Power Budget:	**Not Specified**
Power Budget Margin:	**N/A**
Junction Temperature:	**125.0°C**
Thermal Margin:	-473.1°C (-40.3 W)
Effective ϑJA:	11.5°C/W
Power supplied to off-chip devices:	0 W
Confidence level:	Low

Launch Power Constraint Advisor to find and fix invalid switching activity

On-Chip Power



utilization

Name	Slice LUTs (53200)	Slice Registers (106400)	DSPs (220)	Bonded IOB (200)	BUFGCTRL (32)
nr_divider_top	548	808	40	98	1
stage3 ($paramod$379ecd25f1665ab0d213a7b716ca7b69b1369608\nr_iteration_stage)	125	194	9	0	0
stage2 ($paramod$379ecd25f1665ab0d213a7b716ca7b69b1369608\nr_iteration_stage)	125	194	9	0	0
stage1 ($paramod$379ecd25f1665ab0d213a7b716ca7b69b1369608\nr_iteration_stage)	125	194	9	0	0
stage0 ($paramod$379ecd25f1665ab0d213a7b716ca7b69b1369608\nr_iteration_stage)	125	194	9	0	0

Sky water 130nm

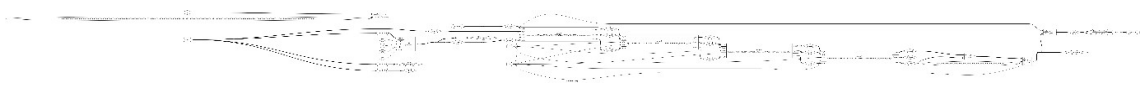
Area before

```
=== design hierarchy ===

   nr_divider_top                    1
     $paramod$379ecd25f1665ab0d213a7b716ca7b69b1369608\nr_iteration_stage      4

   Number of wires:              56591
   Number of wire bits:         378310
   Number of public wires:          83
   Number of public wire bits:    2478
   Number of ports:                 37
   Number of port bits:            874
   Number of memories:               1
   Number of memory bits:          512
   Number of processes:              0
   Number of cells:              48520
     $meminit                        1
     $memrd_v2                       1
     sky130_fd_sc_hd__a211o_1        6
     sky130_fd_sc_hd__a211oi_1      34
     sky130_fd_sc_hd__a21bo_1       27
     sky130_fd_sc_hd__a21boi_0      83
     sky130_fd_sc_hd__a21o_1       221
```

Netlist before



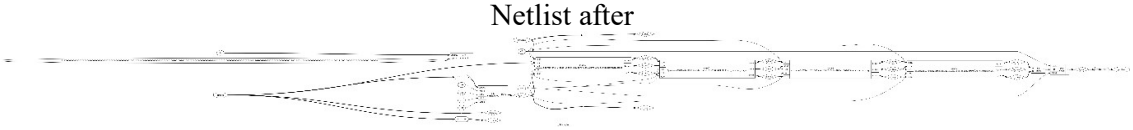
Area after

```
=== design hierarchy ===

   nr_divider_top                    1
     $paramod$379ecd25f1665ab0d213a7b716ca7b69b1369608\nr_iteration_stage      4

   Number of wires:              57227
   Number of wire bits:         376165
   Number of public wires:          83
   Number of public wire bits:    2478
   Number of ports:                 37
   Number of port bits:            874
   Number of memories:               1
   Number of memory bits:          512
   Number of processes:              0
   Number of cells:              48168
     $meminit                        1
     $memrd_v2                       1
     sky130_fd_sc_hd__a2111o_1       4
     sky130_fd_sc_hd__a211o_1        6
     sky130_fd_sc_hd__a211oi_1      22
     sky130_fd_sc_hd__a21bo_1       19
     sky130_fd_sc_hd__a21boi_0      91
```

Netlist after



GF180MCU

Area before

```
=== design hierarchy ===

   nr_divider_top                    1
     $paramod$379ecd25f1665ab0d213a7b716ca7b69b1369608\nr_iteration_stage      4

   Number of wires:             113675
   Number of wire bits:         116541
   Number of public wires:          83
   Number of public wire bits:    2462
   Number of ports:                 37
   Number of port bits:            874
   Number of memories:               0
   Number of memory bits:            0
   Number of processes:              0
   Number of cells:              54461
     gf180mcu_fd_sc_mcu7t5v0__and2_1   3533
     gf180mcu_fd_sc_mcu7t5v0__and3_1    284
     gf180mcu_fd_sc_mcu7t5v0__and4_1    152
     gf180mcu_fd_sc_mcu7t5v0__aoi211_1     39
     gf180mcu_fd_sc_mcu7t5v0__aoi21_1   3872
     gf180mcu_fd_sc_mcu7t5v0__aoi221_1      4
     gf180mcu_fd_sc_mcu7t5v0__aoi22_1    565
     gf180mcu_fd_sc_mcu7t5v0__clkinv_1   3976
     gf180mcu_fd_sc_mcu7t5v0__dffq_1    992
     gf180mcu_fd_sc_mcu7t5v0__mux2_1      1
     gf180mcu_fd_sc_mcu7t5v0__nand2_1  13837
     gf180mcu_fd_sc_mcu7t5v0__nand3_1    355
```

Netlist before

```
digraph "nr_divider_top" {
label="nr_divider_top";
rankdir="LR";
remincross=true;
n10877 [ shape=diamond, label="a_bus0", color="black",
fontcolor="black"];
n10878 [ shape=diamond, label="a_bus1", color="black",
fontcolor="black"];
n10879 [ shape=diamond, label="a_bus2", color="black",
fontcolor="black"];
n10880 [ shape=diamond, label="a_bus3", color="black",
fontcolor="black"];
n10881 [ shape=diamond, label="a_bus4", color="black",
fontcolor="black"];
n10882 [ shape=diamond, label="b_bus0", color="black",
fontcolor="black"];
n10883 [ shape=diamond, label="b_bus1", color="black",
fontcolor="black"];
n10884 [ shape=diamond, label="b_bus2", color="black",
```

Area after

```
=== design hierarchy ===

   nr_divider_top                    1
     $paramod$379ecd25f1665ab0d213a7b716ca7b69b1369608\nr_iteration_stage      4

   Number of wires:              53535
   Number of wire bits:          55914
   Number of public wires:          83
   Number of public wire bits:    2462
   Number of ports:                 37
   Number of port bits:            874
   Number of memories:               0
   Number of memory bits:            0
   Number of processes:              0
   Number of cells:              54461
     gf180mcu_fd_sc_mcu7t5v0__and2_1   3533
     gf180mcu_fd_sc_mcu7t5v0__and3_1    284
     gf180mcu_fd_sc_mcu7t5v0__and4_1    152
     gf180mcu_fd_sc_mcu7t5v0__aoi211_1     39
     gf180mcu_fd_sc_mcu7t5v0__aoi21_1   3872
     gf180mcu_fd_sc_mcu7t5v0__aoi221_1      4
     gf180mcu_fd_sc_mcu7t5v0__aoi22_1    565
     gf180mcu_fd_sc_mcu7t5v0__clkinv_1   3976
     gf180mcu_fd_sc_mcu7t5v0__dffq_1    992
     gf180mcu_fd_sc_mcu7t5v0__mux2_1      1
     gf180mcu_fd_sc_mcu7t5v0__nand2_1  13837
     gf180mcu_fd_sc_mcu7t5v0__nand3_1    355
```

Netlist after

```
digraph "nr_divider_top" {
label="nr_divider_top";
rankdir="LR";
remincross=true;
n5129 [ shape=diamond, label="a_bus0", color="black", fontcolor="black"];
n5130 [ shape=diamond, label="a_bus1", color="black", fontcolor="black"];
n5131 [ shape=diamond, label="a_bus2", color="black", fontcolor="black"];
n5132 [ shape=diamond, label="a_bus3", color="black", fontcolor="black"];
n5133 [ shape=diamond, label="a_bus4", color="black", fontcolor="black"];
n5134 [ shape=diamond, label="b_bus0", color="black", fontcolor="black"];
n5135 [ shape=diamond, label="b_bus1", color="black", fontcolor="black"];
n5136 [ shape=diamond, label="b_bus2", color="black", fontcolor="black"];
n5137 [ shape=diamond, label="b_bus3", color="black", fontcolor="black"];
n5138 [ shape=diamond, label="b_bus4", color="black", fontcolor="black"];
n5139 [ shape=octagon, label="clk", color="black", fontcolor="black"];
n5140 [ shape=diamond, label="d_high", color="black", fontcolor="black"];
n5141 [ shape=octagon, label="dividend", color="black",
fontcolor="black"];
n5142 [ shape=octagon, label="divisor", color="black",
```

Conclusion

This lab successfully implemented a 16-bit Newton-Raphson divider and evaluated its performance on a target FPGA platform using Yosys with Xilinx synthesis, as well as on SkyWater 130nm and GF180MCU ASIC technologies. The design was simulated using Icarus Verilog and verified through GTKWave waveform analysis to ensure correct functionality before synthesis. For the FPGA implementation, synthesis was performed with pre- and post-optimization schematics generated to visualize the logic optimization process, while timing and utilization reports confirmed the divider's performance metrics. For the ASIC implementations, both SkyWater 130nm and GF180MCU processes were targeted, with area reports showing the physical footprint before and after optimization, and netlist comparisons demonstrating the effectiveness of synthesis optimizations across different process nodes. The results show that optimization techniques applied during synthesis significantly improved implementation metrics across all platforms, reducing area and enhancing performance characteristics. Ultimately, the Newton-Raphson 16-bit divider successfully meets the design requirements across all target platforms, with the iterative algorithm efficiently computing division operations while maintaining acceptable area utilization and timing closure for FPGA, SkyWater 130nm, and GF180MCU technologies.